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MILITARY SPECIFICATION

GENERATOR, HYDROGEN ML-539/TM

1. SCOPE

1.1 Scope.-This specification covers the design details and performance characteristics of Generator, Hydrogen ML-539/TM, hereinafter referred to as the ML-539/TM.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

O-A-445	Ammonia, Anhydrous (Compressed)
W-B-101	Batteries and Dry Cells
VV-F-800	Fuel Oil, Diesel
VV-F-815	Fuel Oil, Burner
PPP-D-729	Drums, Metal, 55-Gallon (for Shipment of Noncorrosive Material)

Military

MIL-V-173	Varnish, Moisture- and Fungus-Resistant, for the Treatment of Communications, Electronic and Associated Electrical Equipment
MIL-C-1283	Can, Gasoline, Military, 5-Gallon
MIL-C-3767/4	Connector, Plug, Electrical (Power, Three-wire, Grounding Type), Type UP121M
MIL-S-4043	Steel, Corrosion-Resisting (Extra Low Carbon Type 304) Plate, Sheet and Strip
MIL-C-4150	Case, Carrying and Storage, Shock and Waterproof
MIL-E-4970	Environmental Testing, Ground Support Equipment, General Specification for
MIL-C-5015	Connectors, Electric, AN Type

FSC 3655

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MIL-P-5425	Plastic, Sheet, Acrylic, Heat Resistant
MIL-S-6721	Steel, Corrosion- and Heat-Resistant (18-8 Stabilized) Plate, Sheet, and Strip (for Aircraft Applications)
MIL-C-7413	Couplings, Quick Disconnect, Automatic Shut-off
MIL-N-7513	Nomenclature Assignment and Nameplate Approval, Contractors Method for Obtaining
MIL-S-7720	Steel, Corrosion-Resistant (18-8) Bars and Forging Stock (Aircraft Quality)
MIL-T-8506	Tubing, Steel, Corrosion-Resistant (304) Annealed
MIL-T-8606	Tubing, Steel, Corrosion-Resistant (18-8 Stabilized)
MIL-C-11733	Cylinders, Compressed Gas (Anhydrous Ammonia, ICC-4A480 or ICC-4AA480)
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-C-18959	Cable, Power, Electrical, Polychloroprene Sheathed
MIL-R-26474	Reliability Requirements for Production Ground Electronic Equipment
MIL-D-70327	Drawings, Engineering and Associated Lists

STANDARDS

Federal

No. 595	Colors

Military

MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Specifications and Standards; Order of Precedence for the Selection of
MIL-STD-803	Human Engineering Criteria for Aircraft, Missile, and Space Systems, Ground Support

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.-The following documents form a part of

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this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

National Bureau of Standards

Handbook H28 - Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

3. REQUIREMENTS

3.1 Preproduction.-This specification makes provisions for preproduction testing.

3.2 Components.-The ML-539/TM shall consist of the following major components:

Item No.	Quantity	Description	See Requirement
1	1 each	Frame	3.8.1
2	1 each	Cabinet	3.8.2
3	1 each	Liquid ammonia receiving assembly	3.8.3.1
4	1 each	Single-pass counterflow heat exchanger	3.8.3.2
5	1 each	Vaporizer	3.8.3.3
6	1 each	Pressure regulator, ammonia vapor	3.5.1.1
7	1 each	Pre-heater	3.5.1.1
8	1 each	Retort	3.5
9	1 each	Flowmeter	3.8.3.4.1
10	1 each	Indicating integrator, flow rate	3.8.3.4.2
11	1 each	Batch controller, product gas	3.8.3.4.3
12	1 each	Back pressure regulator, product gas	3.5.1.1
13	1 each	Oil burner	3.6.8
14	1 each	Accessories	3.8.4.

3.3 Selection of specifications and standards.-Specifications and standards for necessary commodities and services not specified herein shall be selected according to MIL-STD-143, except as provided in 3.3.1 and 3.3.2.

3.3.1 Commercial parts.-Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standard parts. In any case, commercial utility parts, such as screws, bolts, nuts, cotter pins, having suitable properties

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may be used provided:

a. They can be replaced by the standard parts (MS or AN) without alteration.

b. The corresponding standard part numbers are referenced in the parts list and, when practical, on the contractor's drawings.

3.3.2 Standard parts.-With the exception outlined in 3.3.1, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their proper MS or AN designation.

3.4 Materials

3.4.1 Metals.-Unfinished metal parts used in fabrication of the ML-539/TM shall be of a corrosion-resistant type. A corrosion-resistant metal part is considered to be one which does not impair in any manner, through direct or indirect effects of its corrosion, the operation of the equipment in any environment likely to be encountered in shipment, storage, and normal service. Metal parts which are subject to corrosion shall be finished in accordance with MIL-F-14072 for Type I exposure. Exterior surfaces, finished in accordance with MIL-F-14072 in preparation for painting, shall be painted with Paint Number 26329 of FED STD No. 595.

3.4.1.1 Dissimilar metals.-Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-F-14072.

3.4.1.2 Metal parts of ML-539/TM process system.-The retort shall be fabricated of corrosion- and heat-resistant nickel alloy with a composition conforming to the following:

Alloy Constituent	Percent Composition by Weight
Nickel	70 (minimum)
Chromium	11 to 15
Iron	10 (maximum)
Manganese	1.0 (maximum)
Silicon	0.5 (maximum)
Carbon	0.15 (maximum).

With the exception of the retort, parts in contact with ammonia

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or the product gas at temperatures higher than 1000 degrees Fahrenheit (F) shall be constructed of steel. (Figure 1 and Table 1). The steel shall be of a stock form appropriate to the fabrication method. It shall have properties of corrosion resistance equal to, or exceeding, the types conforming to the following military specifications: MIL-S-6721, MIL-S-7720, MIL-T-8606, MIL-S-4043, or MIL-T-8506. Parts in contact with the ammonia or product gas at ambient temperature shall be constructed of a corrosion-resistant material. Anodized aluminum or protected cast steel is suitable for this purpose.

3.4.1.3 Structural metal.-The ML-539/TM frame shall be fabricated of SAE 1010 or SAE 1020 carbon steel. These two grades of steel have the following chemical compositions:

	Percent Composition by Weight	
Chemical Constituent	SAE 1010	SAE 1020
Carbon	0.08 to 0.13	0.18 to 0.23
Manganese	0.30 to 0.60	0.30 to 0.60
Phosphorous	0.040 maximum	0.040 maximum
Sulphur	0.050 maximum	0.050 maximum
Iron	remainder	remainder.

3.4.1.4 Cabinet panels.-Sheet metal used to fabricate the ML-539/TM cabinet panels shall conform to grades SAE 1010 or SAE 1020 cold rolled sheet steel.

3.4.1.5 Plastic materials.-Plastic material used to fabricate protective cases for electrical components, such as relay switches, shall conform to MIL-P-5425.

3.4.1.6 Fungus-resistant materials.-Whenever practical, materials which are nutrients for fungi shall not be used. When a fungus-nutrient type of material is used and not hermetically sealed, it shall be treated with one of the following fungicidal agents appropriate to the type of material: Salicylanalide, Copper-8-quinolate, Paranitrophenol, Pyridyl mercuric acetate, Phenyl mercuric oleate, or 2,2' methylenebis (4 chlorophenol).

3.4.1.6.1 Fungus control of electrical elements.-Switches and electrical circuit elements shall be coated with fungistatic varnish conforming to MIL-V-173, Type I.

3.5 Design.-The ML-539/TM shall be a controlled process system which produces a hydrogen and nitrogen gas mixture by a steady

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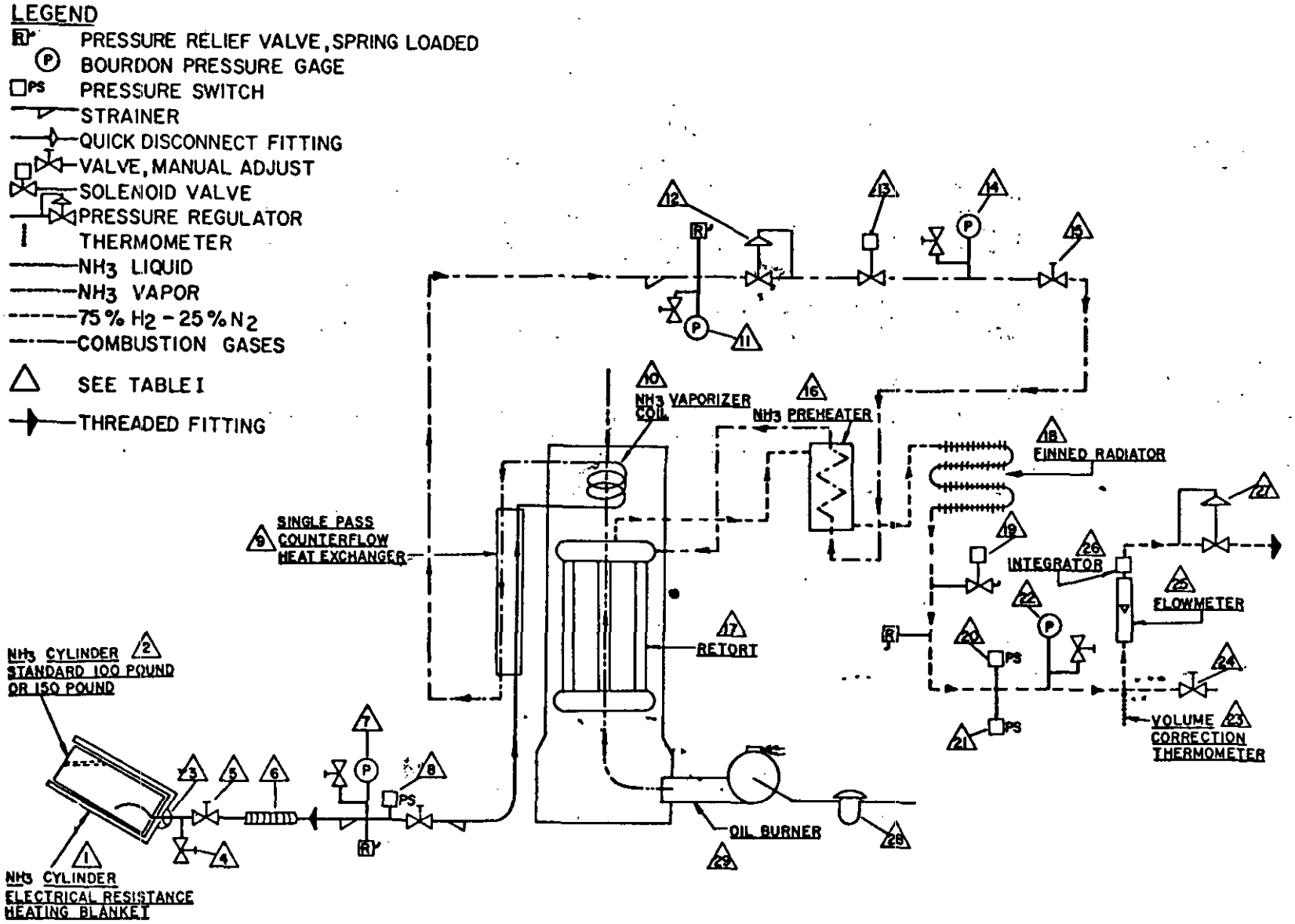


FIGURE I. ML-539/TM PROCESS SYSTEM

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TABLE I

Elements of ML-539/TM Process System

Item No.	Item Description
1	NH_3 Cylinder Electrical Resistance Heating Blanket
2	NH_3 Cylinder, Standard 100-pound or 150-pound
3	Ammonia Cylinder Yoke
4	Vent Valve
5	NH_3 Shut-off Valve
6	Flexible NH_3 Hose
7	NH_3 Pressure Gage
8	NH_3 High-pressure Switch
9	Single Pass, Counterflow Heat Exchanger
10	NH_3 Vaporizer Coil
11	Vaporizer Pressure Gage
12	Pressure Regulator
13	NH_3 Main Flow Solenoid Valve
14	Regulated NH_3 Pressure Gage
15	NH_3 Main Flow Throttling Valve
16	NH_3 Pre-heater
17	Retort
18	Finned Radiator
19	Pressure Relief Solenoid Valve
20	NH_3 Main Flow Solenoid Valve Pressure Switch
21	Safety Pressure Switch
22	Outlet Pressure Gage
23	Volume Correction Thermometer
24	Sampling Valve
25	Flowmeter
26	Integrator
27	Back Pressure Regulator
28	Fuel Oil Filter
29	Oil Burner

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flow process through a retort in which an endothermic, platinum catalyzed dissociation of anhydrous ammonia occurs. The energy for the heat of dissociation of the ammonia shall be supplied to the retort by the combustion gases of an oil burner. The chemical process in the retort shall conform to the following reaction:

$$2NH_3 \xrightarrow{\text{Heat and platinum catalyst}} 3H_2 + N_2$$

where NH_3 is anhydrous ammonia, H_2 is hydrogen, and N_2 is nitrogen. In the above chemical reaction, the reactant and the products are in the gaseous state.

3.5.1 Major equipment characteristics.-The ML-539/TM shall consist of the necessary parts, assemblies, instrumentation, and accessories to perform the functions specified herein. The overall weight of the ML-539/TM, without accessories, shall not exceed 900 pounds.

3.5.1.1 Process system.-The process system of the ML-539/TM shall receive liquid ammonia, pass the liquid through a heat exchanger, and then through a vaporizer. The vaporizer shall receive energy for the heat of vaporization of ammonia from the combustion gases of the ML-539/TM oil burner. The ammonia vapor leaving the vaporizer shall make a return pass through the heat exchanger counterflow to the incoming liquid ammonia, thus cooling the vapor. The cooled ammonia vapor shall then pass through a pressure regulator, a throttling valve, and a pre-heater, and enter the retort where the catalyzed ammonia dissociation occurs. The hydrogen and nitrogen product gases leaving the retort at an elevated temperature shall make a return pass through the pre-heater, thus providing the necessary energy to raise the temperature of the incoming ammonia vapor flowing through the pre-heater. The product gas, partially cooled by an energy exchange in the pre-heater, shall be further cooled by flowing through a finned radiator. The cooled product gas leaving the finned radiator shall then pass through a flowmeter with an indicating integrator unit which shall measure the flow rate and determine the volume of the product gas. After the product gas flow rate has been measured, it shall pass through a back pressure regulator which controls the delivery pressure of the product gas.

3.6 Performance characteristics

3.6.1 Product gas composition.-The ML-539/TM shall produce a consistent composition gas mixture of three parts by volume of

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hydrogen, one part of nitrogen, and 0.04 or less parts of ammonia from a feedstock of liquid anhydrous ammonia. Ammonia shall conform to O-A-445.

3.6.2 Product gas flow rate.-The equipment shall deliver product gas at any preset flow rate between 60 and 400 standard cubic feet per hour. One cubic foot of gas, measured at one atmosphere pressure and 32 degrees Fahrenheit (F), is one standard cubic foot. The maximum flow rate variation shall be plus or minus 2 percent per minute of the preset flow rate. The maximum delivery pressure shall be 8 pounds per square inch gage (psig).

3.6.3 Product gas batch control.-When the ML-539/TM is producing gas at a rate between 60 and 400 standard cubic feet per hour, it shall be capable of delivering any preset quantity (batch) of product gas between 1 and 400 standard cubic feet. It shall automatically stop the gas flow when this quantity has been delivered.

3.6.4 Product gas temperature.-When the ML-539/TM is operating at an ambient temperature of 68 degrees F, the maximum product gas temperature shall be 105 degrees F.

3.6.5 Ammonia cylinder temperature control.-The temperature of ammonia contained within the standard cylinder shall be maintained at 50 degrees F, plus or minus 5 degrees F. This shall be accomplished by using a thermostatically-controlled, electrical resistance heating blanket for any ambient operating temperature below 50 degrees F.

3.6.6 Standby operation.-The ML-539/TM shall be capable of providing standby service by maintaining a retort temperature of 1000 degrees F for periods of at least 5 hours while there is no gas production.

3.6.7 Reliability.-A reliability analysis shall be performed in accordance with MIL-R-26474. The mean operating time between failures (MTBF) shall be at least 300 hours. This requirement shall be demonstrated to a confidence level of 90 percent. A failure shall be defined as the complete cessation of performance or degradation of performance below the levels required in this specification.

3.6.7.1 MTBF.-The MTBF shall be determined under the following conditions:

a. Operation on a cycle of 5 hours continuous operation, of which at least 3 hours shall be at rated capacity, followed by 3

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hours down time.

b. Ambient outdoor temperature.

c. Types of fuels specified to be changed weekly.

3.6.8 Fuel, oil burner.-The equipment shall obtain the necessary heat energy for the retort and vaporizer of the process system by combustion of any of the following fuels:

a. Fuel oil, Diesel (VV-F-800, Grades DF-A, DF-1, or DF-2)

b. Fuel oil, Burner (VV-F-815, Grades F.S. No. 1 or F.S. No. 2)

c. JP-4.

The rate of fuel consumption shall not exceed 3/4 gallon per hour.

3.7 Service conditions

3.7.1 Temperature.-The equipment shall operate at any temperature within a range of minus 40 degrees F to plus 160 degrees F, unsheltered, in all weather conditions. The high temperature shall be considered as that resulting from an ambient air temperature of 125 degrees F and solar heating of 35 degrees F.

3.7.2 Humidity.-The equipment shall withstand exposure to relative humidity up to 95 percent in combination with temperatures of 122 degrees F.

3.7.3 Barometric pressure.-The equipment shall operate at any barometric pressure within the range of 31 to 20.58 inches of mercury. The equipment shall withstand pressure within the range of 31 to 13.75 inches of mercury during nonoperating or storage conditions.

3.7.4 Salt atmosphere.-The equipment shall withstand salt atmosphere conditions encountered in coastal regions, sea locations, and during ocean transportation.

3.7.5 Insects and fungi.-The equipment shall withstand insects and fungi encountered in the tropics.

3.7.6 Rainfall.-The equipment shall withstand continuous rainfall at the rate of 4 inches per hour for a period of 2 hours, accompanied by winds of 20 knots.

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3.7.7 Sand and dust.-The equipment shall withstand sand and dust encountered in desert regions.

3.7.8 Mechanical.-The equipment shall be so designed and constructed that no fixed part shall become loose, no movable part or permanently set adjustment shall shift its setting or position, and no degradation in performance below the minima of this specification shall result from the mechanical influences below.

3.7.8.1 Shock.-The equipment shall withstand repeated 10 g shocks of 11, plus or minus 1 milliseconds duration.

3.7.8.2 Vibration.-The equipment shall withstand vibration testing for Category A equipment, as derined in MIL-E-4970, Category V.

3.7.9 Electrical.-The equipment shall operate with electrical power (for instrumentation, pump work, and minor heating) obtained from 105 to 125 volts, 50 or 60 cycles per second (cps), single phase power source, with a maximum power consumption of 2-3/4 kilowatts.

3.8 Details of assemblies and parts

3.8.1 Frame assembly.-The frame shall be fabricated of steel (3.4.1.3) angles and channels, joined by welding, and shall provide structural support for the elements of the ML-539/TM process system and cabinet. It shall not fracture or undergo plastic deformation as a result of normal loads experienced in handling during shipment and in tactical service. Skid mounts shall be attached to the base of the frame to provide sufficient clearance when the equipment rests on a level surface for handling by a fork lift truck. Lifting eyes shall be provided to facilitate equipment handling by the use of a sling and crane hoist. A sling, fabricated of nylon and metal fasteners, shall be provided for lifting the equipment. The skids, or an attachment thereto, shall be designed for bolting the ML-539/TM to the surface on which the equipment rests.

3.8.2 The ML-539/TM cabinet.-The cabinet (3.4.1.4) shall enclose the process system equipment with the necessary provisions for ducting air to the oil burner, exhausting the combustion gases, and ventilating the finned radiator. The combustion gas exhaust flue, which shall vent at the top of the cabinet, shall be provided with a conical rain hood. The conical hood shall be easily removed. A flat rain shield shall be provided to close the exhaust

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gas vent when the equipment is in shipment and storage. Cabinet panels, in conjunction with the frame, shall provide structural support for mounting instruments, switches, manual controls, and receptacles for fittings of the electrical, liquid ammonia, and the product gas lines. A recessed, weatherproof enclosure shall be provided as an integral part of the cabinet. This enclosure shall afford a centralized mounting space for the electrical components (relay switches, etc.). The electrical relay enclosure shall be approximately 18 inches in width, 12 inches in height, and 8 inches in depth. Access to the relay enclosure shall be through a hinged door in the instrument panel. The lower 18-inch side of the door shall be hinged. The hinge shall be of sufficient strength to maintain proper alignment and clearances of the door. A slide-type stop mechanism shall be used to limit the rotation of the relay enclosure door about the hinge to 90 degrees and to prevent damage to the hinge and any instrumentation attached to the door. The door shall be equipped with a resilient, rubber grommet to effect a weather-proof seal when the door is locked in position. A rain shield and drain shall be positioned immediately above the relay enclosure door to prevent the collection of water along the seal at the top edge of the door. Whenever practical, cabinet panels shall be hinged, or easily removed, to provide immediate access to the process system for maintenance or repair. Fasteners used to hold removable panels in position shall be 1/4 turn, quick-disconnect screws. Hinged doors and panels shall be provided with a positive latching or locking device. No friction or magnetic holding device shall be used. The top of the cabinet shall extend 2 inches beyond the instrument panel surface, with the extension slanting downward approximately 15 degrees from the horizontal. The overall dimensions of the frame and cabinet assembly, including the skid mounts and the conical hood, shall not exceed 35 inches in width, 48 inches in depth, and 66 inches in height.

3.8.3 ML-539/TM process system.-The process system shall consist of the basic elements described in Figure 1 and Table 1, exclusive of the standard commercial ammonia cylinder. The liquid ammonia shall be supplied to the system from 100-pound and 150-pound cylinders. The 100-pound ammonia cylinder shall have a 12-inch inside diameter conforming to MIL-C-11733. The 150-pound ammonia cylinder shall have a 14-1/2-inch inside diameter conforming to MIL-C-11733.

3.8.3.1 Liquid ammonia receiving assembly.-The process system shall have an assembly to receive the liquid ammonia from the standard cylinder. The assembly shall have a yoke (an adapter) capable of being attached to the outlet of the standard ammonia cylinder by means of a threaded fitting or other suitable device.

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The liquid ammonia shall pass through the yoke to a manually-operated ammonia shut-off valve. The ammonia flow passage between the inlet to the yoke and the ammonia shut-off valve shall have provision for venting to the atmosphere through a manually-operated vent valve. Liquid ammonia passing through the shut-off valve shall flow into the process system within the ML-539/TM cabinet through a 10-foot long flexible ammonia hose. This hose shall be fabricated of corrosion-resistant steel and shall consist of a flexible, bellows construction hose protected by a corrosion-resistant armor of separate, interlocking segments. The flexible ammonia hose shall be connected to the shut-off valve by threaded fittings. Connection of the hose to the rigid tubular line within the ML-539/TM cabinet shall be made with threaded fittings.

3.8.3.1.1 Electrical resistance heating blanket.-The ammonia shall be forced into the process system by its vapor pressure. A thermostatically controlled, electrical resistance heating blanket shall be provided for the ammonia cylinder to maintain adequate vapor pressure of the liquid ammonia.

3.8.3.1.2 Ammonia cylinder rack.-A rack shall be provided to hold the ammonia cylinder. When the rack rests on a horizontal mounting surface, the longitudinal axis of the ammonia cylinder, mounted on the rack, shall be inclined at a 5-degree angle with the horizontal. The rack shall support the cylinder by two surfaces which conform to the external curvature of the cylinder for an arc-length which subtends an angle of 75 degrees. The supporting surfaces shall be 2 inches in width and spaced 30 inches center to center, oriented perpendicular to the longitudinal axis of the cylinder. The rack shall have two sets of supporting surfaces mounted back to back. One set of the supporting surfaces shall conform to the curvature of the 150-pound ammonia cylinder and the other to the curvature of the 100-pound ammonia cylinder (3.8.3). Provision shall be made to bolt the rack to a mounting surface. Webbing straps and any necessary hardware fittings shall also be provided for securing the cylinder to the rack.

3.8.3.2 Single-pass counterflow heat exchanger.-The heat exchanger shall consist of two concentric tubes and the necessary fittings to pass the liquid ammonia through the smaller tube and the vaporized ammonia through the annular space between the tubes. The flow of the liquid and the vapor shall be in opposite directions. The heat exchanger shall provide sufficient cooling of the ammonia vapor to protect the pressure regulator from overheating in any operating condition.

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3.8.3.3 Vaporizer coil.-The vaporizer shall consist of a helically coiled tube designed to extract sufficient energy from the oil burner combustion gases to evaporate all ammonia passing through the vaporizer.

3.8.3.4 Instrumentation and controls.-Instruments and controls specified herein, and such additional items considered necessary by the contractor for efficient and safe operation of the ML-539/TM, shall be centrally located on a single instrument panel. Provision shall be made to illuminate the instruments. Wherever practical, the instruments shall be illuminated internally. In addition to scale divisions, instrument dials and scales shall be color coded to indicate that the system function being measured is within tolerance, marginal, or beyond tolerance limits causing an inoperative or unsafe condition. The requirements of MIL-STD-803 shall apply for instrument and control arrangement, labeling, scale divisions, and color markings on instrument faces, color marking of transilluminated indicators, and instrument illumination.

3.8.3.4.1 Flowmeter.-The ML-539/TM shall be equipped with a flowmeter capable of measuring and indicating the volume flow of the product gas in standard cubic feet per hour units. The flowmeter shall be calibrated for the following conditions:

a. One atmosphere (14.696 pounds per square inch absolute) ambient atmospheric pressure.

b. Product gas delivery pressure of 8.0 psig.

c. Product gas composition conforming to the requirements of 3.5.

d. Product gas temperature at the flowmeter of 68 degrees F.

A dial thermometer (Figure 1) shall be used to measure and indicate the product gas temperature at the flowmeter. The flowmeter measurement requires a correction factor for any condition of calibration varying from those stated above. The correction factor for gas temperature variation shall be indicated with the corresponding gas temperature on the face of the dial thermometer, based on the conditions stated in paragraphs a, b, and c. A flowmeter calibration chart shall be provided to show correction factor versus product gas temperature. Flowmeter measurement correction required for ambient atmospheric pressures less than one atmosphere shall be shown on the calibration chart. The parameter for the additional

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curves, ambient atmospheric pressure, shall be stated in inches of mercury. The flowmeter calibration chart shall provide a correction factor for any product gas temperature encountered in normal operation (3.7.1) and for any ambient atmospheric pressure corresponding to equipment operation at altitudes between mean sea level and 10,000 feet above mean sea level. The chart shall be sealed between transparent plastic sheets. The accuracy of the flow rate measurement shall be consistent with the accuracy requirement of the product gas volume measurement (3.8.3.4.2).

3.8.3.4.2 Indicating integrator.-An indicating integrator shall be provided which shall determine from the flow rate measurement (3.8.3.4.1) the volume of product gas which has passed through the flowmeter. The volume shall then be reported on a 6-digit counter. The integrator shall sense the flow rate a minimum of 60 times per minute, at equal time increments, and actuate the counter by one electrical signal for each standard cubic foot of product gas without the need of a multiplier in the circuitry. The volume measurement (standard cubic foot units) shall be accurate to within plus or minus 4 percent for gas produced at any rate between 60 and 400 standard cubic feet per hour. Provision shall be made for checking the accuracy of the integrator when there is no gas production. The integrator unit shall be provided with electrical contacts and have means to actuate them. The integrator shall close the contacts once for each standard cubic foot of product gas measured. The electrical contact device shall be in addition to the signaling device which operates the 6-digit counter. The electrical contacts shall be selected for service in a line which operates at approximately 5 amperes, 105 to 125 volts, and 50 or 60 cps, single phase.

3.8.3.4.3 Batch process control.-A controlling device shall be provided which regulates the process system to produce a preset volume (batch) of product gas. This controller shall regulate the production of 1 to 400 standard cubic foot batches for any integral number of volume units within this range to an accuracy of plus or minus 4 percent. The batch controller shall employ the volume measurement of the flowmeter and indicating integrator instruments. In a normal gas production run, the batch controller shall stop the gas flow of the ML-539/TM after the preset volume has been delivered. It shall reset itself to zero and provide the necessary controls to maintain the elements of the ML-539/TM process system at operating temperature. The batch process control shall be capable of being manually reset to zero after a partial batch has been delivered. The batch controller shall have provision for isolation from the control circuit to allow continuous gas production.

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3.8.3.4.4 Pressure gages.-A bourdon-type pressure gage with a bezel sealed face shall be provided to measure the operating pressure at each point of the ML-539/TM process system indicated by Figure 1 and Table 1. The line to each pressure gage shall be provided with a tee-fitting and manually-operated valve which vents to the atmosphere. The valve shall have a standard threaded fitting to facilitate adaptation of a standardized pressure gage for checking and calibrating the bourdon-type pressure gages in the equipment. The valves and fittings shall be readily accessible by maintenance personnel.

3.8.3.4.5 Pressure regulation within the retort.-A pressure regulation device shall be provided to maintain a constant pressure within the retort. This pressure regulation shall be accomplished with a minimum of gas waste.

3.8.3.4.6 Pressure relief valves.-The ML-539/TM process system shall be provided with a spring-loaded type pressure relief valve at each point indicated in Figure 1. These valves shall be easily removed from the system for inspection, cleaning, or replacement.

3.8.3.4.7 Temperature measurement and control of the retort.-The retort of the ML-539/TM shall be provided with a temperature measuring, indicating, and controlling device. The sensing element of the measuring device shall be a chromel-alumel thermocouple with ceramic insulation. The measuring device shall control the oil burner in order to maintain a preset temperature on the retort. Provision shall be made for manually adjusting the retort temperature for standby or operating conditions. This adjustment shall be made by a knob or similar mechanism externally mounted to the cabinet on the instrument panel. The temperature indicating device shall utilize transilluminated indicators. These indicators shall show whether the retort is at the preset temperature, above this temperature, or below it. The temperature measurement shall be accurate to within plus or minus 1 percent. The controlling device shall prevent discharge of product gas when the retort is at a temperature outside the range of operating temperature specified by the contractor.

3.8.3.4.8 Safety alarm system.-The ML-539/TM shall stop operating, sound an alarm, and by means of a lighted indicator on the instrument panel, show that any one or combination of the following malfunctions or failures exists:

a. Retort rupture.

b. Overheat of retort.

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c. Thermocouple burnout or disconnection of thermocouple leads.

d. Flowmeter rupture.

e. Electrical power supply failure.

f. Oil burner failure.

The visual and sound alarm system contained within the ML-539/TM shall be paralleled by a remote alarm system with 150 feet of weatherproof cable. The sound alarm system shall have provisions for silencing the alarm once it has been detected by the operator. The lighted indicators shall remain illuminated until the malfunction or failure has been corrected. The alarm system shall use batteries conforming to W-B-101 where circuitry independent of the power source is required.

3.8.3.4.8.1 Simulation of retort and flowmeter rupture.-The gas sampling valve of the ML-539/TM (Figure 1) shall be of sufficient size that when fully opened, the pressure drop caused in the line shall simulate rupture of the flowmeter or retort.

3.8.3.4.9 Elapsed-time indicator.-The oil burner motor of the ML-539/TM shall be provided with an elapsed-time indicator. This device shall totalize and indicate the operating time of the oil burner motor.

3.8.4 Accessories

3.8.4.1 Product gas delivery hose.-A 50-foot plastic hose, resistant to fuels, oils, water, and ammonia shall be provided to deliver the product gas. The hose shall be red in color and have a 3/4-inch inside diameter. The wall thickness shall be selected for operating pressures of 10 psig and maximum pressures of 50 psig. The physical and chemical properties of the hose shall assure its successful use throughout the range of environmental operating conditions (3.7). It shall resist kinking. One end of the hose shall be equipped with a threaded fitting to mate the product gas outlet connection of the ML-539/TM. The other end shall have an adapter for Nozzle, Meteorological Balloon Inflation, ML-196. The hose shall have an electrical conductor internally contained within its wall thickness. The electrical conductor shall be connected to the adapter and threaded fitting in a manner to assure an electrostatic potential difference of zero between the balloon inflation nozzle and the ML-539/TM process system. A ball-type valve shall be provided at the delivery end of the hose.

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3.8.4.2 Fuel receiving assembly.-The ML-539/TM shall be provided with a 15-foot fuel delivery hose and the required fittings to receive fuel from either a 5-gallon gasoline can conforming to MIL-C-1283 or from a 55-gallon metal drum conforming to PPP-D-729, Type I. A 5-gallon can shall be provided (3.13). Vents shall be provided for each type of container to prevent entry of water during rainfall. The fuel hose shall be coupled to the fuel line within the ML-539/TM cabinet by a quick-disconnect fitting conforming to MIL-C-7413. Preceding the oil burner connection, two filters shall be provided in the fuel line. The first filter shall be a clear plastic bowl type for collection of water. It shall have provision for draining the water without necessitating removal of the bowl. The second shall be a sediment filter with a replaceable cotton waste cartridge. The fuel line within the cabinet shall have provision for electrical resistance heating.

3.8.4.3 Electrical power supply cable.-A 50-foot electrical power supply cable for the ML-539/TM shall be provided with quick-disconnect fittings. The cable shall conform to MIL-C-18959.

3.8.4.4 Electrical quick-disconnect fittings.-The quick-disconnect fittings used for the power supply cable and electrical resistance heating blanket shall conform to MIL-C-5015.

3.8.4.5 Electrical grounding rod.-The ML-539/TM shall be furnished with an electrical grounding rod having provisions for electrical connection to the frame of the ML-539/TM.

3.8.4.6 ML-539/TM equipment case.-An equipment case shall be designed and provided by the contractor to contain and protect the accessories and loose parts from damage and loss in shipment and storage. The case shall conform to MIL-C-4150.

3.8.4.7 Product gas analysis.-Plastic bottles, graduated containers, tubing, and fittings required for a volumetric product gas analysis of NH_3 content shall be provided. The plastic material shall be sufficiently rigid to assure accuracy to within 1 percent in volumetric measurement. The plastic shall be resistant to acidic and basic solutions and liquid hydrocarbon fuels, solvents, and lubricants. A 6-liquid ounce container of methyl red indicator and a pipette with bulb dropper shall be provided.

3.8.4.8 Ammonia leak detectors.-One hundred combustible, sulphur impregnated sticks used for ammonia leak detection, shall be provided with each ML-539/TM.

3.8.4.9 Electrical receptacle.-An electrical receptacle with a conduit box of weatherproof design shall be provided as a

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convenience power outlet. The receptacle shall have a threaded cover plate. An anchored chain attached to this plate shall be provided to prevent its loss. The receptacle shall receive parallel bladed plugs with grounding provisions, such as the UP121M plug, conforming to MIL-C-3767/4. A 10-ampere fuse shall be connected in series with the circuitry of the receptacle. The electrical line conditions at this outlet shall be 105 to 125 volts, 50 or 60 cps, single phase (3.7.9). The electrical power rating shall be clearly labeled at the receptacle. The receptacle and its fuse shall be readily accessible.

3.9 Maintenance.-The ML-539/TM shall be designed and constructed to allow maintenance with tools and equipment normally available commercially by maintenance personnel with a minimum of training. Its design and construction shall allow ease of assembly and disassembly of any parts or components which require periodic servicing, replacement, or calibration. In order to facilitate maintenance, threaded fittings and parts shall conform to National Bureau of Standards Handbook H-28. Human engineering design considerations for maintenance of the ML-539/TM shall conform to the applicable requirements of MIL-STD-803.

3.10 Interchangeability.-All parts having the same manufacturer's part number shall be electrically, mechanically, and functionally interchangeable. The drawing number requirements of MIL-D-70327 shall govern changes in the manufacturer's part numbers.

3.11 Identification of product.-Nomenclature assignments and approval of nameplate identification shall be obtained in accordance with MIL-N-7513. The equipment shall be marked in accordance with MIL-STD-130.

3.12 Government-loaned property.-When the contract or purchase order so provides, the Government will loan the following to the contractor upon his request:

Quantity per unit	Description	Federal Stock No.
1 each	Nozzle, Meteorological Balloon Inflation, ML-196.	6660-526-7846

3.13 Government-furnished property.-When the contract or purchase order so provides, the Government will furnish the following to the contractor upon his request for use in making the ML-539/TM:

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Quantity per unit	Description	Federal Stock No.
1 each	Can, Gasoline, Military, 5-gallon	7240-222-3088.

3.14 Workmanship.-This equipment shall be constructed in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes and defects, accuracy of dimensions, markings, and freedom of parts from burrs and sharp edges. Metal shall form clean, smooth, and unstrained bends and turns where such shaping is necessary.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests.-The inspection and testing of the ML-539/TM shall be of the following classifications:

a. Acceptance tests (4.3)

b. Preproduction tests (4.6).

4.3 Acceptance tests.-Each ML-539/TM shall be subjected to all tests described under paragraph 4.5, Test methods.

4.4 Test conditions.-Unless otherwise specified, all acceptance tests shall be made at an atmospheric pressure between 28 and 32 inches of mercury and at a temperature of 77 degrees F, plus or minus 18 degrees F.

4.5 Test methods

4.5.1 Examination of product.-Each ML-539/TM shall be examined to determine strict compliance with the requirements of this specification. Particular attention shall be given to the following: visual defects in material and workmanship, dimensions, marking,

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packaging, and packing.

4.5.2 Performance tests

4.5.2.1 The ML-539/TM shall be operated through the following test cycle:

a. Start and warm up to a retort standby temperature of 1000 degrees F; maintain at the standby condition for 30 minutes with no gas production.

b. Bring the retort to operating temperature. Start gas production; adjust the product gas flow rate to 400 standard cubic feet per hour.

c. Perform the tests specified in 4.5.2.1.2, 4.5.2.1.3, and 4.5.2.1.4.

d. The ML-539/TM shall then be shut down.

4.5.2.1.1 Number of test cycles.-The test cycle shall be performed three times. The retort temperature shall be allowed to cool below 400 degrees F between test cycles. A different grade of fuel oil for the burner shall be used in each test cycle. The following types of fuel shall be used:

a. Fuel Oil, Burner, Grade F.S., No. 2, of VV-F-815.

b. Fuel Oil, Diesel, Grade DF-1, of VV-F-800.

c. JP-4.

The fuels shall be used in the sequence indicated for the three test cycles.

4.5.2.1.2 Test cycle number one

a. Before the retort has reached operating temperature, open the gas outlet valve. No product gas shall flow since the unit has not reached operating temperature.

b. After gas production has begun (adjusted to 400 standard cubic feet per hour), lower the high temperature index of the retort temperature controller to below the operating temperature. This condition shall simulate overheating of the retort. The product gas flow shall stop, and an alarm shall sound. Silence

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the alarm; adjust the temperature controller for normal operation, and start gas production.

c. Disconnect one thermocouple lead at a time to simulate thermocouple burn out. The temperature controller shall actuate controls to stop gas flow, shut down the unit, and sound an alarm. Silence the alarm; connect the thermocouple lead, and start gas production after each shut down.

d. While gas is being produced, disconnect electrical power cable to the generator. The ML-539/TM shall shut down, and a battery-operated bell alarm shall sound. Silence the alarm; connect the power cable, and start gas production.

e. Shut off gas flow at the end of delivery hose. The product gas flow from the process system and the ammonia flow to the retort shall stop. A pressure switch shall operate the solenoid relief valve to prevent occurrence of excessive pressure in the system between the retort and product gas outlet resulting from continuing dissociation of NH_3 trapped in the retort.

4.5.2.1.3 Test cycle number two

a. Fully open the gas sampling valve to simulate retort or flowmeter rupture. The product gas flow shall stop, and an alarm shall sound. Silence the alarm; close the gas sampling valve, and start the gas production.

b. While the gas is being produced, check each bourdon-type pressure gage with a standardized gage. The bourdon gages shall be accurate to within the smallest subdivision of their dials.

4.5.2.1.4 Test cycle number three.-After the product gas delivery has been started and adjusted to a flow rate of 400 standard cubic feet per hour, a volumetric gas analysis shall be performed by withdrawing a product gas sample from the gas sampling valve. The gas flow shall be stopped. A 200 standard cubic foot batch shall be delivered using the batch controller. During the delivery of this batch, the volume of product gas shall be measured by connecting the output to a separate, calibrated, totalizing gas meter of known accuracy. The volume of the entire batch shall be measured. The time for delivery shall be measured by a stop watch to verify the 400 standard cubic foot per hour flow rate. The product gas flow shall be stopped by the batch controller after the 200 standard cubic feet have been delivered. The retort shall be maintained at operating temperature. The fuel flow rate shall be measured during the delivery of the batch by the use of a stop watch and

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burette. The minimum volume of fuel measured shall be 500 milliliters.

4.6 Preproduction testing

4.6.1 Preproduction test sample.-The preproduction test sample shall consist of an ML-539/TM representative of a production item. It shall be tested at a laboratory approved by the procuring activity, or when so stated in the contract, at the contractor's plant under the supervision of the procuring activity.

4.6.2 Preproduction tests.-The test sample (4.6.1) shall be subjected to the following series of tests:

a. All tests described under 4.5, Test methods.

b. The vibration test as specified in MIL-E-4970, Procedure V.

c. The rain test as specified in MIL-E-4970.

d. The test sample shall be placed in a cold chamber at minus 40 degrees F with its fuel supply, liquid ammonia supply, and power supply connected. In order to approach thermal equilibrium, the test sample shall be left in the chamber with all electrical power off for a minimum of 4 hours prior to starting. After the cooling period, the electrical power shall be turned on for the required initial electrical heating. The ML-539/TM test sample shall then be operated continuously in the minus 40 degrees F environment for 48 hours through the following cycle:

(1) Two and one-half hours of gas production at 400 standard cubic feet per hour.

(2) One and one-half hours of standby operation.

The product gas shall be ducted from the cold chamber via the product gas hose and any additional fittings and tubing required. Provisions shall be made to duct the combustion gases of the oil burner from the chamber. The fuel supplied to the oil burner shall conform to Grade DF-A of VV-F-800. The product gas composition shall be sampled and measured once during each 2-1/2-hour gas production period. The NH_3 content shall not exceed 1 percent by volume. The total volume of product gas shall be measured by a calibrated, totalizing meter of known accuracy which is separate from the volume measuring device in the ML-539/TM. The product gas volume thus measured shall be recorded at the end of each 2-1/2-hour gas production period. The calibrated meter and the ML-539/TM

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product gas volume measurements shall not vary by more than plus or minus 4 percent. The electrical power consumption shall be measured and recorded throughout the test and shall not exceed 2-3/4 kilowatts at any time. The total fuel consumption in 48 hours of operation shall not exceed 36 gallons. The test sample shall experience no malfunction requiring repair or shut down of the equipment during this period.

e. The test described above (4.6.2 d) shall be repeated in a chamber maintained at 120 degrees F. The fuel used in the oil burner shall be grade DF-2 of VV-F-800, in exception to the above procedure.

f. The shock test, as specified in MIL-E-4970, Procedure I, shall apply for an uncased, upright-mounted test sample. The test sample, packaged for shipment and delivery, shall be subjected to the shock test specified in MIL-E-4970, Procedure II.

g. The sand and dust tests with sand and dust velocity at 2300, plus or minus 500 feet per minute, as specified in MIL-E-4970, Procedure I.

h. The salt fog tests, as specified in MIL-E-4970.

i. The fungus resistance tests as specified in MIL-E-4970, Procedure I.

j. Low pressure test as specified in MIL-E-4970.

k. Reliability test (see requirement, 3.6.7).

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with the instructions of the procuring activity.

6. NOTES

6.1 Intended use.-The Generator, Hydrogen ML-539/TM is intended for use as a tactical hydrogen generator for the inflation of meteorological sounding balloons.

6.2 Ordering data.-Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Whether the item in 3.12 shall be Government-loaned property.

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c. Whether the item in 3.13 shall be Government-furnished property.

d. Number of preproduction samples, the place of inspection, and instructions for inspection records and their availability.

e. Instructions for preparation for delivery.

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MATERIAL PROCURED UNDER A

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A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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